⑥ Variable _ Data structure

☐ Creating a Java file

Extension = .Java

☐ Boilerplate code :-

Class → Public class JavaBasics {

Main function → Public static void main (string args[]){

[Note :→ In Java Name of class & Name of file must be same]

```
    - - -
    }
}
```

☐ Output in Java :-

System.out. Print ("Hello world") ; — Statement terminator

Output

→ Program :

```
public class JavaBasics {
    Public static void main (String args[]) {
        System.out. println ("Hello world!");
        System.out. println ("Hello world! ");
        system.out. println ("Hello  world!");
    }
}
```

[Println is used too because of space in line.]

(Note :- we can use \n for seperate line)

→ output in Java.

① Print → system. out. print ("Hello world!") ; → string

② Println → system. out. println ("H W!");

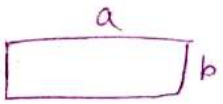
③ "\n" → \n

Print A Pattern

```
* * * *      S.o.p ln("* * * *");
* * *        S.o.p. ln("* * *");
* *          S.o.p. ln("* *");
*            S.o.p. ln("*");
```

Variables in Java

$a = 10$, $b = 5$



$2 * (a + b)$

Literal Variable

In Java :-

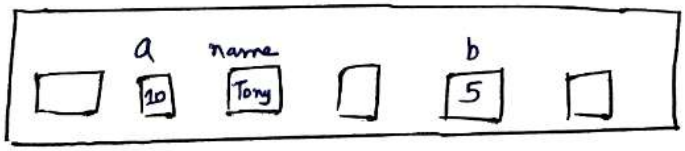
```
int a = 10;
int b = 5;
```

Identifier

a, b, c ....
Java Basics
main, print, print ln

} ~~Identi~~ Identifiers

• Memory



Let
string name = "Tony";

→ In memory there are so many blocks in organized way.

~~Program~~

```
int a = 10;
int b = 5;
system.out.print ln (a);
system.out.print ln (b);
String name = "Tony";
system.out.print ln (name);
```

Double cote is not because we have to print their value.

Output! -

10
5
Tony

```
a = 50;
system.out.print ln (a);
```

50

☐ Data types in Java

Primitive	Non primitive
byte	String
Short	Array
char	class
boolean	object
int	Interface
long	
float	
double	

CPP, Python, Java ⟶ Typed language

Note → All the datatypes must be start from lowercase (small) word.

• Program

```
public class JavaBasics{
    public static void main (String args[]){
        byte b = 8;
        System.out.println (b);
        char ch = 'a';
        system.out.println (ch);
        boolean var = false;
        float price = 10.5;
        //double
        int number = 25;
        //long.
        Short n = 240;
    }
}
```

→ Size of Data Types :-

∴ 1 Byte = 8 Bit

1 Bit { [0] [1] } either 0 or 1.

- byte — 1 byte — [-128 to 127] 256 values
- short — 2 bytes — [ ]
- char — 2 bytes — ['a' to 'z', 'A' to 'Z', '@' '?' ',']
- boolean — 1 byte — true, false
- int — 4 bytes — -2B to +2B
- long — 8 bytes — ✓ more big
- float — 4 bytes — decimal value
- double — 8 bytes — Big decimal value.

□ Sum of a & b

```
int a = 10;
int b = 5;
int sum = a + b;
syso (sum)
```

Program

```
{
  int a = 10;
  int b = 5;
  int sum = a+b;
  System.out.print(sum);
}
```

Output = 15

Comments in Java

// Hi there! → Single line

/*
Hi there
*/ → multiline comment

Input in Java :-

- next
- nextLine
- nextInt
- nextByte
- nextFloat
- nextDouble
- nextBoolean
- nextShort
- nextLong

Code

```
import java.util.*;

public class JavaBasics {
  public static void main(String args[]){
    Scanner sc = new Scanner (System.in);
    // String input = sc.next();
    // System.out.println(input);

    // String name = sc.nextLine();
    // System.out.println(name);

    int number = sc.nextInt();
    System.out.println(number);

    float price = sc.nextFloat();
    System.out.println(Price);
```

Sum of a & b

Scanner →

```
int a = sc.nextInt();
int b = sc.nextInt();
int sum = a+b;

Sys o (sum);
```

- Product of A & B.

Code.

```
import java.util.*;

public class JavaBasics {
    public static void main (String args[]) {
        Scanner sc = new Scanner (System.in);
        int a = sc.nextInt();
        int b = sc.nextInt();

        int product = a*b;
        System.out.println (product);
    }
}
```

- Area of circle:

$3.14, \frac{22}{7}$

Area = π * rad * rad

Code

```
- - - -
    - - - {
    Scanner sc = new Scanner (System.in);
    float rad = sc.nextFloat();
    float area = 3.14f * rad * rad
    System.out.println (area)
}
```

(To tell Java that it is float value.)

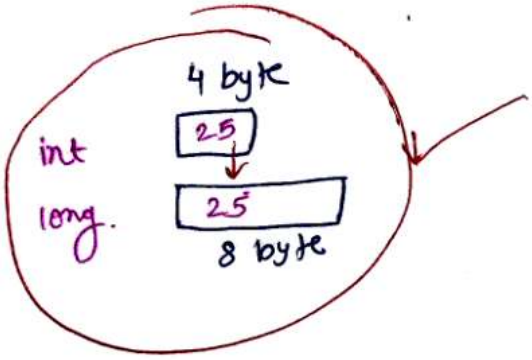
- Type conversion :- (widning, implicit conversion)

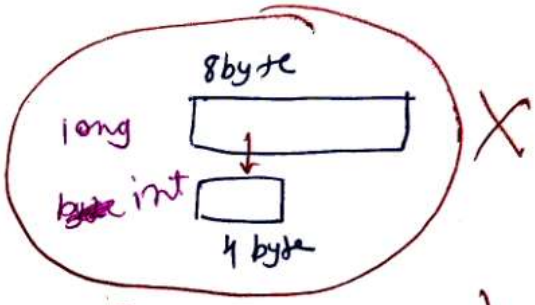
→ Conversion happens when :-

a. type compatible

b. destination type > Source type

byte → Short → int → float → long → double





(Lossy conversion)

- Type casting :- (narrowing, explicit conversion)

int marks = (int) (99.99f)

Code

```
- - - {
- - - {
    char ch = 'a';
    char ch2 = 'b';
    int number = ch;
    int number2 = ch2;
    Sysoln (number);
    syso ln(number 2);
  }
}
```

Output

97 (a)
98 (b)

Type Promotion in Expressions:-

1. Java automatically promotes each byte, short, or char operand to int when evaluating an expression.

2. If one operand is long, float or double the whole expression is promoted to long, float or double respectively.

```
a + b * c / e
↙   ↓   ↓   ↘
int float char long
```

Code

```
..... {
public ... {
  int a = 10;
  float b = 20.25f;
  long c = 25;
  double d = 30;
  double ans = a + b + c + d;
  System.O.Pln (ans);
```

Output:

85.25

1. byte, short, char $\xrightarrow{\text{convert}}$ int
2. long, float, double $\longrightarrow$ big data types

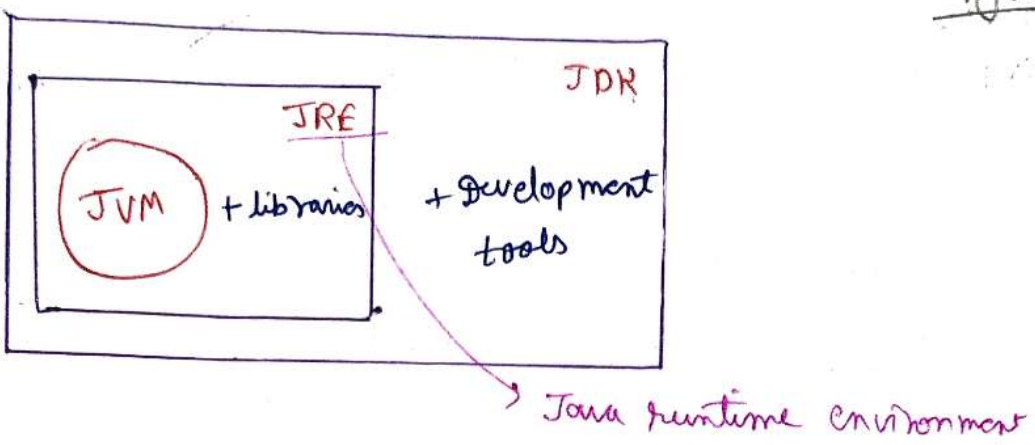
```
// wrong.
byte b = 5;
b = b * 2;
```

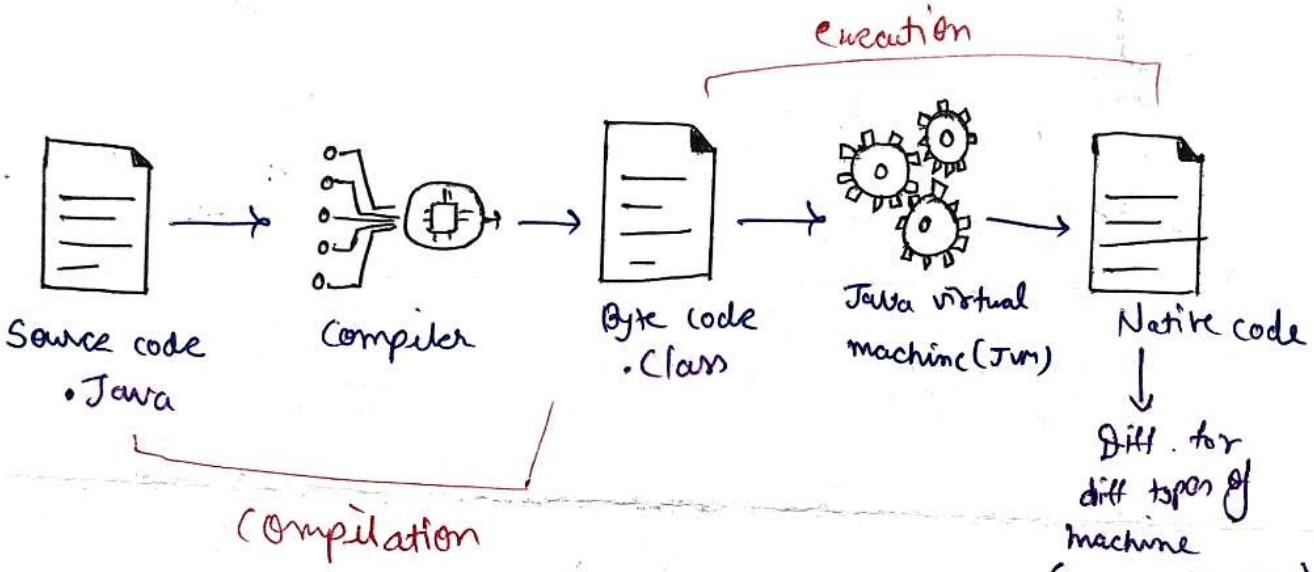
expression

$= 98 \times 2 = 196$ (range of byte = -128 to +127)

```
// right
byte b = 5;
b = (byte)(b * 2);
```

▫ How does Java code run :-





➤ Practice Questions

(1) In a Program, input 3 number: A, B & C. You have to output the average of these 3 numbers!

```
... {
    public static void main (String args[]) {
        Scanner Sc = new Scanner (System.in);

        int A = Sc.next();
        int B = Sc.next();
        int C = Sc.next();
        int Avg = (A+B+C) / 3;
    } System.out.print (Avg);
}
```

Output

```
4
2
6
---
4
```

(2) In a Program, input the Side of a Square. You have to output the area of the square. Pg 19

```
import. java.util.*;
- - - - -
public static void main (String args[]){
        Scanner SC = new scanner (System.in);
        int side = Sc.nextInt ();
        int area = side * side;
        System.o.pln ("area of the square is:" + area);
    }
}
```

Input = 4
output = area of the square is: 16

---

(3) Enter cost of 3 items from the user (using float data type) - a Pencil, a pen and an eraser. You have to output the total cost of the items back to the user as their bill. (with 18 % GST)

```
import - - -
- - - -
        Scanner sc = new Scanner (System.in);
        float Pencil = sc.nextFloat ();
        float Pen = sc.nextFloat ();
        float Eraser = sc.nextFloat ();

        float Total = Pen + Pencil + Eraser;
        System.out.println ("Total :" + Total);

        float Newtotal = Total + (0.18f * Total);
        System.out.println ("Bill with GST:" + Newtotal);
    }
}
```

Input :- 5, 25, 10:
output :- Total : 40.0
Bill with GST :. 47.2

(4) What will be the type of result in the following code

Pg. 20

```
byte b = 4;
char c = 'a';
short s = 512;
int i = 1000;
float f = 3.14f;
double d = 99.9954;

result = (f * b) + (i % c) - (d * s);
```

Ans:- In the mentioned code, the result variable will be of double type because of type conversion.

(Q.5) (Advanced) ~~what~~ will this code following statement will give any error in Java?

```
int $ = 24;
```

Ans: No, the statement will not give any error. Names of variable are called identifiers in Java. Identifiers can start with any aphabet or underscore ("_") or doller ("$").

According to the rule the given variable name is a valid identifier.